| 卷册检索号<br>SEARCH NO. OF THE VOLUME |
|---|
| MLHVDC-LCS-ECI-DD-CA-0218<br>30-B003701S-T0218 |

第 1 页 共 1 页
PAGE OF

# 图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detailed Design DESIGN STAGE

建筑 专业 第 02 卷 第 18 册
Architecture SPECIALITY VOLUME BOOK

卷册名称 专用品库建筑图(第一次修改)
Volume name Special materials storage(First revision)

| | |
|---|---|
| 批 准<br>APPROVED BY | |
| 审 核<br>REVIEWED BY | |
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|---|---|---|---|---|---|---|
| 1 | 30-B003701S-T0218-001 | 专用品库建筑图 设计说明<br>Special materials storage Description of design | B | 1 | | |
| 2 | 30-B003701S-T0218-002 | 专用品库建筑图 设计说明 (英语版)<br>Special materials storage Description of design(English version) | B | 1 | | |
| 3 | 30-B003701S-T0218-003 | 专用品库建筑图<br>Special materials storage | B | 1 | | |

注： 本次升版图纸为001、002、003图。
The revised drawings are 001、002、003.

## 建筑设计说明

一、项目概况：

项目名称：巴基斯坦默蒂亚里—拉合尔±660kV 直流输电工程/拉合尔±660kV 换流站

建设单位：国家电网/中国电力技术装备有限公司

| 单体名称 | 层数 | 火灾危险性分类 | 建筑高度 | 建筑面积 | 室内外高差 |
|---|---|---|---|---|---|
| 专用品库 | 1 | 戊 | 5.35 | 28.41 | 0.45 |

结构形式：框架 结构使用年限：50 年 屋面防水等级：I 级 耐火等级：二级

二、建筑施工注意事项：

1. 本工程建筑和装饰用料的选用，规格及色彩应满足设计要求。建筑饰面及装修材料规格和色彩应满足设计要求。
2. 土建施工队、安装施工队装饰等各专项施工队伍间、各专业间均应密切配合，施工或安装前先应全面清楚了解有关工种设计图纸内容要求（包括基础结构部份）。发现设计问题或现场有变化情况时，应及时与设计人员联系，协商确定解决方法，进而确保工程进度与施工质量。
3. 本项目工程建筑关键部分用料，如不锈钢、塑料制品、防水卷材、建筑密封膏、外墙饰面材料、防火防护门窗、吊顶、厕所隔断、内外装饰用材、油漆和涂料等的规格和质量等要求，须经监理确认无误，以确保工程质量。
4. 建筑平面所注尺寸均为结构尺寸，所注门窗尺寸均为洞口尺寸，墙体定位均为墙体中心至轴线的尺寸。
5. 凡与工程配套的门窗等专项设备与产品，按大样图选型。由制造厂商或分包施工单位提出预留洞、预埋件的资料或图纸，并经业主及设计方确认，后方可施工。其中，门窗洞口尺寸以现场土建完成的洞口为准，须实地测量，
6. 辅助用房内(储藏室、工具间等)不可大量存贮易燃物品。

三、建筑消防设计：

单体间间距均大于 9m，满足消防间距要求。每个单体为独立的防火分区。

出入口不少于 2 个。采用自然排烟，排烟窗面积大于地面面积 2%。

四、节能设计：

1.外窗（含玻璃门透明部分）断桥隔热铝合金窗框，窗玻璃采用普通钢化玻璃 6+12A+6。

2.屋面采用正置式，保温板(XPS)厚度定为 50 厚，屋面挤塑聚苯板（XPS）耐火等级 B1，容重要求 30 kN/m³。

3. 室内遮阳采用活动遮阳（土建不施工，装修时东，南，西三面外窗内侧安装遮光帘）。、

五、建筑施工说明

1.墙体

1.1 概述

1.1.1 本项目砌块采用当地标准机制粘土烧结砖，砖长度为 230，砖及砂浆强度详见结施。

1.1.2 从基础向上砌筑的墙体在标高-0.060 处设置 1 层 20 厚 1:3 水泥砂浆(掺 5%防水剂)为防潮层。(该部位有结构圈梁则可以取消墙体防潮层)

1.1.3 外粉刷必须设置分格缝。

1.1.4 设置通长现浇钢筋混凝土窗台梁，高度不宜小于 120mm，配筋详结施；其他层在窗台标高处应设置通长现浇钢筋混凝土板带。窗口底部混凝土板带应做成里高外低；房屋两端顶层砌体沿高度方向应设置间隔不大于 1.3m 的现浇钢筋混凝土板带。板带的配筋及混凝土强度详结施。

1.1.5 当洞宽大于 2m 时，洞口两侧设置混凝土构造柱（并与雨篷梁或框架梁同时浇筑），配筋详结施；当洞宽小于 2m 时，在洞口两侧的下部混凝土板带上，设置止水坎，其高度为 1～2 皮砖的厚度，宽度不小于 120mm。

1.1.6 对门框与柱距离小于 300mm 的门垛及小于 360mm 窗间墙，宜采用钢筋混凝土浇筑。

1.1.7 宽度大于 300mm 的预留洞口，设钢筋混凝土过梁，尺寸同门窗洞口过梁。预留的门窗洞口边需采用混凝土构造柱（与雨篷梁或框架梁同时浇筑），配筋详结施。

1.1.8 墙体内的埋管密集区域，采用混凝土浇筑。

1.1.9 在两种不同基体交接处，应采用钢丝网抹灰或耐碱玻纤网布聚合物砂浆加强带进行处理，加强带与各基体的搭接宽度不应小于 150mm。顶层粉刷砂浆中宜掺入抗裂纤维。

1.1.10 室内墙、柱面、门窗的阳角，在 2000 高度内均做 1:2 水泥砂浆护角线，每侧宽度不小于 50。

1.1.11 外墙窗台、窗楣、雨篷、阳台压顶和突出腰线等上面应做流水坡度，且不应小于 25，下口应设滴水槽，深度为 10mm，宽度为 10-12mm，距水平板外缘 30mm 处或离墙面 30mm 处设置断水口，滴水槽内涂水柏油。

1.1.12 设置通长现浇钢筋混凝土窗台梁高 200mm，配筋详结施。房屋两端顶层应设置通长现浇混凝土板带，间隔 1.3m。板带高 80mm，配筋详结施。

1.1.15 内墙墙面粉刷层满铺抗裂钢丝网。外墙与成品电缆桥接触部分为混凝土实心砖砌筑。如遇混凝土墙面，先刷素水泥浆（加 3%建筑胶）一道，满批腻子打光，再做外墙饰面。勒脚墙砖吸水率 E≤3%，破坏强度≥1300N.可根据实际砖铺贴尺寸，微调勒脚高度，尽量采用整砖。

1.1.16 墙体无约束的端部必须增设构造柱。

1.1.17 内墙乳胶漆耐洗刷性（次）≥1000，VOC 含量≤100g/L；瓷砖吸水率不大于 6%，破坏强度不小于 600N；

1.1.18 室内宜台贴黑色人造石，人造石突出墙面约 20mm，沿口倒圆角。窗台板与墙面结合处采用白色硅胶封闭；

1.2 墙体构造做法

1.2.1 外墙 1 做法(用于勒脚处)：

1:1 水泥砂浆掺粘结剂(或专用粘结剂)粘贴深灰色面砖(配套填缝剂勾缝)

20 厚 1:3 水泥砂浆找平

砌块墙体作基层处理

1.2.2 外墙 2 做法(用于勒脚以上)：

丙烯酸外墙涂料面漆两遍

满涂底漆一道

外墙腻子批平

20 厚 1:3 水泥砂浆找平

不同材料间用钢丝网片铺贴

砌块墙体基层处理

2．地面

20 厚 1：3 水泥砂浆找平

150 厚混凝土，含水率<-6%

1.5 厚聚氨酯防水层（两道）

20 厚 1：3 水泥砂浆找平

素水泥浆一道（掺建筑胶）

80 厚混凝土垫层内配双向φ10@250 钢筋网

素土夯实（夯实系数不小于 0.94）2.1.2 楼面（其它房间）

3．屋面工程：

3.1 概述

3.1.1 屋面找平层应设分格缝，分格缝纵横间距不大于 6m，缝宽为 20mm，并嵌填密封材料。

3.1.2 屋面刚性保护层应采用细石防水混凝土，厚度不应小于 50mm，并设置分格缝，其间距不宜大于 6m，缝宽不应大于 30mm，且不小于 12mm。刚性防水层与山墙、女儿墙及突出屋面结构的交接处，应留置伸缩缝，伸缩缝用柔性防水材料填充，并铺设高度、宽度均不小于 250mm 卷材附加层。找坡层坡度为 3%；混凝土内配间距为 100～200mm 钢筋网片，钢筋网片应位于刚性保护层的中上部，且在分隔缝处断开。

3.1.3 屋面有组织排水落水管采用φ110 UPVC 管，落水管离地 1m 高度设置检查口，地面设成品雨水口。

3.1.6 屋面刚性层与女儿墙、山墙之间预留 30mm 伸缩缝，伸缩缝用柔性防水材料填充，并铺设高度、宽度均不小于 250mm 的卷材附加层。

3.1.7 屋面女儿墙、压顶等过长的纵向构件，应沿纵向不大于 3m 设置钢筋混凝土构造柱，女儿墙、压顶粉刷层每隔 3m 及易产生变形开裂部位设分格缝，分格缝宽为 10mm。

3.2 屋面构造做法

3.2.1 保温屋面

50 厚 fc'=28MPa 细石混凝土面层(内配 φ4@100 网片)(分隔缝间距不大于 6m)

20 厚 1：3 水泥砂浆保护层

2 道共 6 厚 SBS 改性沥青卷材错缝搭接（排水沟处增加一道）

土工布一层

50 厚挤塑聚苯泡沫塑料板（XPS）

1.2 厚聚氨酯涂料隔汽层一道

20 厚 1：3 水泥砂浆找平层

最薄处 30 加气混凝土找坡层（排水沟处无层）

现浇混凝土结构屋面板

3.2.2 雨篷

20 厚 1：3 水泥砂浆保护层

总厚度 1.5 厚聚氨酯防水涂料 2 道

1：3 水泥砂浆找坡找平层（向地漏处找坡）

现浇混凝土结构楼板

4．门窗

4.1 铝合金窗型材壁厚必须不小于 1.4mm，门的型材壁厚必须不小于 2mm。

4.2 凡防火门均带闭门器与逃道锁，门扇内外均安装拉手。

4.3 外门构造应开启方便，坚固耐用；手动开启的大门扇应有制动装置，推拉门应有防脱轨的措施；双面弹簧门应在可视高度部分装透明的安全玻璃。

4.4 门立樘位置除有特殊要求者外，弹簧门、平开内门立樘居中；平开外门立樘居中，并应注意有防止雨水渗入的措施。窗立樘居砖墙中心。

4.5 铝合金门窗外框与墙体的缝隙填塞海绵条或玻璃棉毡条分层填塞，缝隙外表留 5-8mm 深的槽口，槽口处嵌填密封膏；

4.6 本施工图门窗立面仅表示分隔及开启方式，门窗制作厂家应在土建完成后，认真核对有关尺寸及数量，并留出安装缝隙，核实无误后方可施工；

4.7 抗风压性能、气密性及水密性满足 GB/T7106-2008 要求。抗风压 4 级：2.5≤P3<3.0，且挠度变形须满足规定；气密性 6 级：1.5≥q1>1.0；水密性 3 级：250≤△P<350；空气隔声性能：GB/T8485-2008 分级指标的 6 级；Rw≥45dB。窗的拼樘料应做抗风压变形计算，与门窗框的拼接应为插接，深度不小于 10mm。

4.8 砖墙上门窗过梁，砼均为 fc'=28MPa。当过梁顶标高与圈梁底标高贴近时，过梁应与圈梁整体浇筑。

4.9 门窗数量及规格要求见门窗数量表，门开启方向详见平面图

4.10 门窗编号由“门窗类别名+门洞宽+洞高”的形式组成。门窗类别名详门窗详图的注释说明

4.11 采用安全玻璃的范围（不小于 5mm 厚度）

（1）面积大于 1.5 平米以及底边离最终装修面小于 500mm 的落地门窗

（2）出入口、门厅等部位

（3）易遭受撞击、冲击而造成人体伤害的其他部位

5.油漆涂料

5.1 除特殊要求之外，一般木质构件做一底二度聚氨酯、树脂调和漆或硝基漆，不露面木构件用专用防腐涂料处理，露面钢铁构件用热镀锌，不露面钢构件做红丹漆二度。

5.2 外墙涂料设分隔线，位置见立面，分隔线宽度 20，采用黑色涂料。

6.其它

6.1 防雷设施:详电施图纸

6.2 建筑图纸中各专业综合留洞示意图后出，电气、水工、暖通等专业留孔大小及位置详见各专业图纸（以各专业图纸为准）。

6.3 任何图纸上有矛盾与疑问之处，要在着手工作之前报请建筑师和工程师澄清，如出现中英文版本不一致的情况，以中文版为准。

| 楼层 | 部位 / 房间名称 | 楼地面做法 | 踢脚 | 墙面 | 顶面 | 备注 |
|---|---|---|---|---|---|---|
| -- | -- | 1.20厚1：3水泥砂浆找平<br>2.150厚混凝土，含水率<=6%<br>3.1.5厚聚氨酯防水层（两道）<br>4.20厚1：3水泥砂浆找平<br>5.素水泥浆一道（掺建筑胶）<br>6.80厚混凝土垫层内配双向φ10@250钢筋网<br>7.素土夯实（夯实系数不小于0.94） | 1.20厚1：3水泥砂浆找平<br>2.素水泥浆一道（掺建筑胶）<br>3.清理基层墙体<br>高度120 | 1.A级白色无机涂料一底二面<br>2.内墙腻子三遍<br>3.6厚1:0.5:2.5混合砂浆抹平<br>4.9厚1:0.5:3混合砂浆扫毛<br>5.墙体非同一材料时接缝处需增加钢丝网片<br>6.素水泥浆1道<br>7.砌块墙体作基层处理 | 1.A级白色无机涂料一底二面<br>2.内墙腻子三遍<br>3.6厚1:0.5:2.5混合砂浆抹平（楼板底平整度误差小于4mm可不做本层）<br>4.素水泥浆1道<br>5.现浇混凝土结构楼板 | |
| | | | | | | |

Reference Drawings

| Drwg. No. | Title |
|---|---|
| 30-H0037015-T0248 | 建筑通用详图 General details of the building |

Revisions

| | | | | |
|---|---|---|---|---|
| A | | 2017.11.30 | 第一版 | First Edition |
| | Name | Date | Notes | |

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 中国电力技术装备有限公司 CHINA ELECTRIC EQUIPMENT GROUP CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detailed Design

M L H Y D C - L C S - E C I - D D - C A - 0 2 1 8 - 0 0 1 B

| | 签名 Name | 日期 Date |
|---|---|---|
| 设计 Designed | | 2019.09.06 |
| 校核 Checked | | 2019.09.06 |
| 审核 Reviewed | | 2019.09.06 |
| 批准 Approved | | 2019.09.06 |
| 比例 Scale | -- | 图号 DWG NO. |

专用品库建筑图

建筑设计说明

Special materials storage

the instruction of architectural design

30-H0037015-T0218-001

版本 REV. B

594X841

## Project Narrative

### 1.General

Project name：±660kV HVDC Project from Matiari to Lahore in Pakistan

Development organization：State Grid / China Electric Power Equipment And Technology CO.,TD

| Name | Story | Height | classification of fire hazards | Area | Altitude intercept |
|---|---|---|---|---|---|
| Special materials storage | 1 | 5.35 | the fifth of the ten Heavenly Stems | 28.41 | 0.45 |

Structural style: frame structure　　Service life of structure: 50 years

Roof waterproofing grade: 1　　Fire resistance rating: grade 2

### 2. Precautions of construction

1) The selection, specifications and color of the construction and decoration materials should meet the design requirements.

2) Different majors of construction teams such as construction team and installation team should cooperate with each other closely. Construction teams should have a comprehensive understanding of the relevant blueprints before construction or installation. When discovering the design mistake or changes during the construction, they should contact the design unit in time to discuss the solution in order to ensure the progress and quality of the project.

3) The specification and quality of material requirements of the key parts such as stainless steel, plastic products, waterproofing materials, construction sealant, facade materials, fire-proof doors and windows, ceilings, toilet partitions, decoration materials, and paintings should beconfirmed by supervisor in order to ensure the quality of the project;

4) The dimensions of the building plan are structural size, and the dimensions of the doors and windows are whole size. Using the dimension from the center to the axis of the wall to locate the wall.

5) Special equipment and products used in this project, such as doors and windows etc. should be selected by detail drawings. The information or blueprints of the reserved holes and embedded parts proposed by manufacturer or construction unit should be confirmed by the clients and the design unit. In addition, dimensions of doors and windows must be confirmed through direct measurement.

6) No large amount of combustible goods can be stored in auxiliary rooms.

### 3: Architectural fireproof design

1) In order to satisfy the requirement of separation distance for fireproof, the distance between two single buildings should be larger than 9m;

2) The building is an individual fire compartment.

3) There are two entrances of building.

4) Natural ventilation is used in the building and the area of the vent windows should be no less than 2% of the net area of each room.

### 4: Energy conservation design

1) The Exterior window frames (including the transparent part of the glass door) are made of bridge heating insulation aluminum alloy, and their windowpanes are made of ordinary tempered glass 6+12A+6;

2) Traditional Insulation System is used for roofs（The insulation layer is below the waterproof layer）, and the width of the insulation board(XPS) is fixed as 50mm; The XPS's fire proof grade is B1; the volume-weight is 30 KN/m3.

3) Indoor sun shading system should apply mobile shading devices.(The shade curtain will be not in the process of construction until the interior decoration is be done. It should be installed inside of the east, south, west window.)

### 5: Construction instructions

#### 1. Wall

1.1 General

1.1.1 The masonry structure of this project uses local clay brick, The brick length is 230 and the standard strength consult structure design drawings.

1.1.2 The walls built from the foundation should set one layer of moisture proof layer at 0.060m level high which is 20mm width and using 1:3 cement mortar (with 5% water-proof agent). (Moisture proof layer can be canceled if beam wall)

1.1.3 Exterior paint must be provided with slit.

1.1.4 The building should be set long cast-in-place concrete sill beam, the height of beam is less than 120mm,the set of steel bar refer to the detail of structure; other layers of floor should be set reinforced concrete slab continuous cast-in-situ belt at the window. The inner concrete slab at the bottom of the window shall be higher than outside, the top masonry at both ends of the house shall be provided with a cast-in-place reinforced concrete slab with an interval of not more than 1.3m along the height.

Structural slab reinforcement and concrete strength consult structure design drawings.

concrete structural column (pouring at time with awning beam or frame beam), the set of steel bar refer to the detail of structure; when the hole width is less than 2m, on both sides of the concrete belt, setting water bucket, the height of 1 to 2 the brick thickness, the width of which is not less than 120mm. Structural slab reinforcement and concrete strength consult structure design drawings.

1.1.6 For doorframe and column which distance between those is less than 300mm or those door stack which is less than 360mm window wall, should use pouring steel concrete.

1.1.7 The reserved hole which width is longer than 300mm should set reinforced concrete beams, which size is the same as the window or door beam. For windows and doors side requires the use of structural concrete column (pouring at time with awning beam or frame beam), the set of steel bar refer to the detail of structure.

1.1.8 Walls are densely embedded pipes in the walls should use pouring concrete.

1.1.9 The strength grade of masonry should be refer to structural design drawings.

1.1.10 The steel bar must be embedded between the balcony and the main part of the building.

1.1.11 Steel wire mesh plastering or alkali resistant fiberglass mesh reinforced polymer mortar reinforcing belt at the juncture of two different materials, and the overlap reinforcing belt shall not be less than anti-cracking fiber.

1.1.12 Indoor wall, doors and windows, cylindrical corner under 2000mm height are 1:2 cement mortar protection line, the width of each side no less than 50mm.

1.1.13 At the part of windowsill, window lintel, awning, balcony top and belt line should set water slope, and should not be less than 25mm, at the bottom there should set drip line slot. The depth is 10mm, the width is 10-12mm, and from the wall 30mm set a water outlet, brushing bit mastic solution into the drip line slot.

1.1.14 On the top and bottom should be set long cast-in-place reinforced concrete beams in the window, the set of steel bar refer to the detail of structure, totally beam is 120mm high. The other layers in the window at the elevation reinforced concrete strip; both ends of the top housing should be set through the concrete slab with interval 1.3m. Structural slab reinforcement and concrete strength consult structure design drawings.

1.1.15 Interior wall's painted layer covered with anti-crack steel wire mesh. The contact part of the external wall and the cable cabinet is made of concrete solid bricks. At the part of concrete wall, there should brush cement paste (3% building glue) at first, then putty, then do exterior finishes. Water absorption rate of plinth's material should be less than 3%(E < 3%), damage strength > 1300N. According to the actual size of tiles, fine-tuning the plinth height, as a whole brick.

1.1.16 Wall must be added with a structural column at its unconstrained end.

1.1.17 Interior wall latex paints scrub resistance (time) should be more than 1000, VOC content is less than 100g/L; tile's water absorption is not more than 6%, the failure strength is not less than 600N.

prominent wall about 20mm, fillet along the corner. The combination of windowsill and wall is sealed with white silica gel.

1.1.19 All lightweight partition wall increase the 200mm height of the concrete door sill, the width the same as walls.

1.2 Walls

1.2.1 Exterior wall 1 (For the plinth):

Dark gray tiles (using relevant joint mixture) are pasted by 1: 1 cement mortar mixed with binder (or special binder);

Smoothed by 20 mm and 1: 3 cement mortars;

Add steel wire mesh to the joints of non-identical materials of the wall;

Treat the masonry walls at the base level;

The decoration layer for surface on interior wall;

1.2.2 Exterior wall 2 (Above the plinth):

Acrylic exterior paint finish twice

Apply a full primer

Smooth the putty on exterior wall;

20 mm and 1: 3 cement;

Wire mesh is used to cover different wall materials

Treat the base level for masonry walls

#### 2. Flooring

20 thick 1:3 cement mortar leveling,

150 thick concrete ,water content < = 6%

1.5 thick polyurethane waterproof layer (two)

20 thick 1:3 cement mortar leveling

Plain cement paste a (mixed building glue)

80 thick concrete cushion with two-way phi within 10 @ 250 mesh

Plain soil compaction (compaction coefficient is not less than 0.94)

#### 3. Roof

3.1 General

3.1.1 Roof leveling layer shall be divided into compartments, the distance between slit spacing is no longer than 6m, width of 30mm, and filling of sealing material. Structural slab reinforcement and concrete strength consult structure design drawings.

3.1.2 Stiff protect layer should be used waterproof fine aggregate concrete, its strength is refer to structural design drawings, the thickness should not be less than 50mm, and set the separation gap, the distance should be less than 6m, width should be less than 30mm, and not less than 12mm. The junction of stiff waterproof layer and gable parapet and prominent roof structure, should be retained expansion joints, expansion joints filled with flexible waterproof material, and the laying of the height and width of not less than 250mm coil additional layer. The slope of slope layer should be 3%; the spacing of the concrete should be 100 to 200mm; the reinforcing steel mesh should be located in the upper part of the stiff protect layer, and it should be broken at the separation joint.

3.1.3 Expansion joints using cast-in-place reinforced concrete plate, the strength grade is not less than fc'=28MPa.; out of the surrounding roof pipe, manhole with a roof structure with cast-in-place reinforced concrete waterproof ring, the height of which is not less than 300mm.

3.1.4 Roofing pipes must be provided with waterproof sleeves and water retaining rings.

3.1.5 Organized roof drainage gutters with ϕ110 UPVC pipe, for 1m from the ground height setting check position. The ground is equipped with product drain well.

3.1.6 Between the stiff layer and parapet or gable, reserved for the 30mm expansion joints, expansion joints filled with flexible waterproof material, with the additional layer which height and width not less than 250mm .

3.1.7 The rainwater drop from the upper roof to the lower roof, the lower roof should set splash pan.

3.1.8 If the longitudinal component such as roofing parapet wall is too long, should set reinforced concrete constructional column, no more than 3m. Along the longitudinal parapet, top layer of paint every 3M and easy to produce deformation and cracking position. Dividing seam, slot width 10mm lattice.

3.2 Roofing construction practices

3.2.1 Insulation roof:

50mm thick fc'=28MPa fine aggregate concrete surface (containing ϕ4@100 meshes）(The distance between joints is not larger than 6m）;

20mm width 1:3 cement mortar protective layer;

Joggled lap 2 layers and totally 6mm width SBS modified asphalt membrane (add one more layer on drainage ditch);

One layer of geotechnical cloth;

50mm XPS;

1.2 mm width Polyurethane coating vapor layer;

20mm 1:3 cement mortar screed-coat;

30mm aerated concrete slope making layer at the thinnest place（no layer on drainage ditch）;

Cast-in-situ concrete structure roof sheathing;

3.2.2 Awning and exterior air condition board：

20 mm width 1:3 cement mortar protective layer;

Totally 1.5 mm width Polyurethane coating （2 layers）;

1：3 cement mortar slope making layer and screed-coat（slope towards the floor drains）;

Cast-in-situ concrete structure floor;

#### 4. Doors and windows

4.1 The thickness of aluminum alloy of window must be no less than 1.4mm; thickness of section bar of the door must be no less than 2mm.

4.2 where the fire door with door closers and channel lock doors are installed with handles inside and outside the window, when the high of the windowsill is less than 800mm, should set up a steel protective railing, door height from the ground floor of not less than 1050mm; the vertical rail spacing is less than 110mm. All protection rails can withstand horizontal thrust no less than 100kg;

4.3 Doors should be easy to open, the door should be durable; fan brake manually open the sliding door should be anti-derailment measures; double spring loaded safety glass doors should be transparent in the visible part of the height.

4.4The door vertical position in the frame with special requirements, spring door, flat open door vertical frame in the middle; flat open door frame vertical center, and should pay attention to prevent rainwater infiltration measures. The window frame in the center of the brick wall.

4.5 Aluminum Alloy frame and wall cracks with sponge or glass wool felt layered tamping, the gap left deep notch appearance 5-8mm, notch block packing paste;

4.6 The construction plan doors and windows facade only means separation and opening, doors and windows manufacturers should be completed after the construction, carefully check the size and quantity, and leave the installation gap, verify that the construction.

4.7 Wind resistance performance, air tightness and water tightness meet the requirements of GB/T7106-2008. Wind resistance level 4: 2.5 < P3 < 3, and the deflection must meet the requirements; tightness level 6: 1.5 Q1 > 1; water tightness Level 3: 250 = Delta P < 350; the sound insulation performance of air quality index: GB/T8485-2008 level 6: Rw = 45dB. The window of the piecing material should be anti-wind pressure deformation calculation, and the window frame should be inserted for stitching, depth of not less than 10mm.

4.8 The concrete of the lintel of Windows and doors on the brick wall is fc '=28MPa.When the elevation of the top of the lintel is close to that of the bottom of the ring beam, the lintel should be poured with the ring beam as a whole.

4.9 Refer to the chart to find the number and the specifications of the doors & windows. Refer to the plan to find the direction of the doors.

4.10 The number of the doors & windows is composed by "category name of door and window + width of its hole + height of its hole ". Lookup the detail drawing to find the its category name.

4.11 Conditions on using shatterproof glass

(1) Landing doors & windows which the area is larger than 1.5 $m^2$ and the bottom edge is less than 500mm from the decorated surface;

(2) Entrances, halls and so on;

(3) Other parts those are vulnerable to impact which can cause human injury.

#### 5. Paint

5.1 Except special requirements, general wooden components use polyurethane, resin mixing paint or nitrocellulose lacquer, hidden wooden artifacts use special anticorrosive paint, apparent steely components use hot galvanizing, hidden steely components use two layers of red lead paint.

5.2 The exterior wall paint is divided into the parting line, and the position is refer to the elevation. The width of the divider is 20mm, which is used the black paint.

#### 6. Else:

6.1 Lightning protection facilities: refer to detail of electrical drawings

6.2 Electrical, hydraulic, HVAC and other specialty's reserve holes, refer to relevant drawings.

6.3 If there are any contradictions on any drawings, contact the designer before construction. In the event of a difference between the Chinese and English versions, the Chinese version shall prevail.

| level | position / name | Floor practice | Skirting line | wall | Ceiling | Remarks |
|---|---|---|---|---|---|---|
| -- | -- | 1.20 thick 1:3 cement mortar leveling,<br>2. 150 thick concrete ,water content < = 6%<br>3.1.5 thick polyurethane waterproof layer (two)<br>4.20 thick 1:3 cement mortar leveling<br>5. Plain cement paste a (mixed building glue)<br>6.80 thick concrete cushion with two-way phi within 10 @ 250 mesh<br>7. Plain soil compaction (compaction coefficient is not less than 0.94) | 1.20 thick 1:3 cement mortar leveling<br>2.Water slurry (construction glue)<br>3.Clean up the base wall<br>The height to 120 | 1.Synthesized resin latex coatings: Once primer, twice pastry;<br>2.Inner wall putty three times<br>3.6 thick 1:0.5:2.5 mixing mortar<br>4.9 thick 1:0.5:3 mixed mortar shred<br>5.When the wall is not the same material, the seam should be increased<br>6.Plain water 1 channel<br>7.Masonry walls are treated at the base level | 1.Synthesized resin latex coatings : Once primer, twice pastry;<br>2.Inner wall putty three times<br>3.6 thick 1:0.5:2.5 mixing mortar(floor flatness error is less than 4mm not to do this layer)<br>4.Water slurry<br>5.Cast-in-place concrete floor slab | |
| | | | | | | |

To decorate a list

Reference Drawings

| Drwg. No. | Title |
|---|---|
| 30-B0037015-T0248 | 建筑通用详图 General details of the building |

| Revisions | | | |
|---|---|---|---|
| A | Name | 2017.11.30 Date | 第一版 First Edition Notes |

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detailed Design

M L H Y D C - L C S - E C I - D D - C A - 0 2 1 8 - 0 0 2 B

| | 签名 Name | 日期 Date |
|---|---|---|
| 设计 Designed | | 2019.09.06 |
| 校核 Checked | | 2019.09.06 |
| 审核 Reviewed | | 2019.09.06 |
| 批准 Approved | | 2019.09.06 |

专用品库建筑图

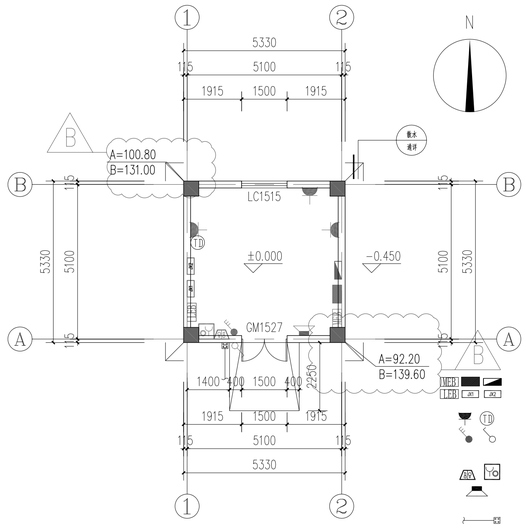
建筑设计说明(英语版)

Special materials storage

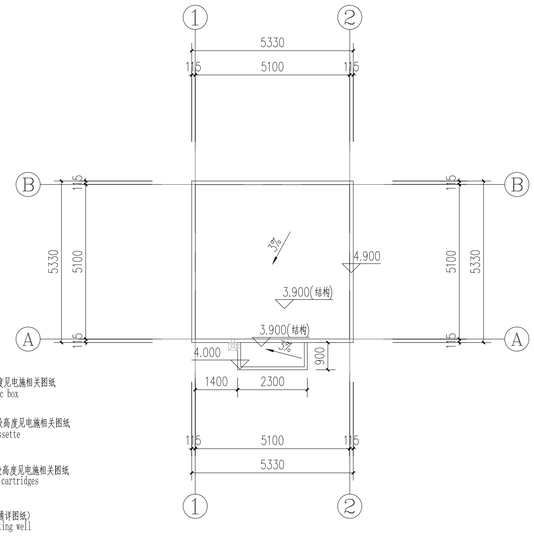
the instruction of architectural design(English version)

比例 Scale: --　图号 DWG NO.: 30-B0037015-T0218-002　版本 REV.: B

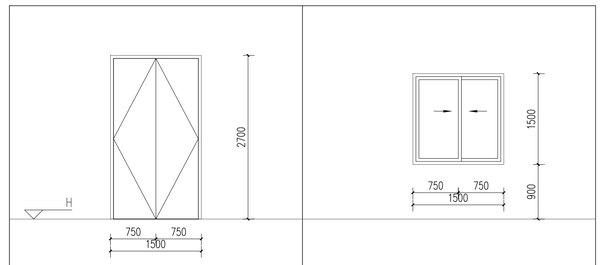
594X841




一层平面图 1:100
First floor plan
S=28.41m²




屋顶平面图 1:100
Roof plan

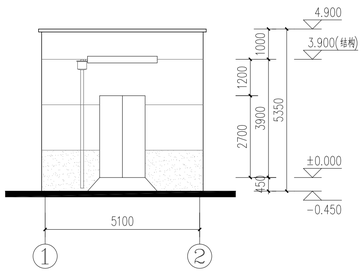



| 编号 number | GM1527 | 洞口尺寸 door hole size | 1500X2700 | 编号 number | LC1515 | 洞口尺寸 door hole size | 1500X1500 |
|---|---|---|---|---|---|---|---|
| 备注 remarks | 钢门 steel door | 数量 number | 1 | 备注 remarks | 铝合金移窗 aluminum alloy moving window | 数量 number | 1 |

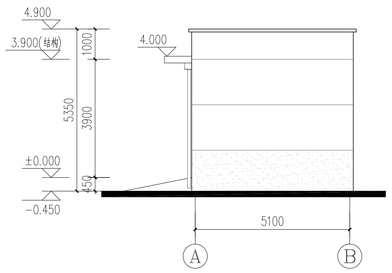
门窗详图 1:50
Windows and Doors detail

外窗断桥隔热铝合金窗框，窗玻璃采用普通钢化玻璃 6+12A+6，所有可开启窗附纱窗。窗外侧设置 30X30 ø2 铁丝网
Outside the window of aluminium alloy of the heat insolation that break the window frame, window using ordinary tempered glass 6 + 12 + 6, all of the windows which can open should install the screen window and out side of the window set 30 x30 ø2 barbed wire.

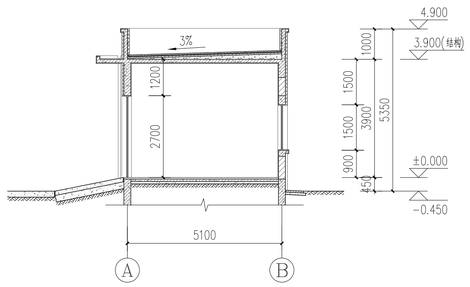
B



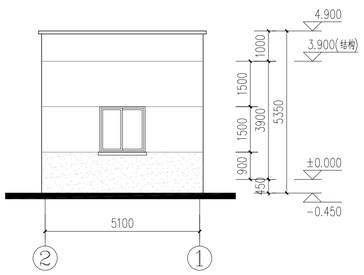

①-②轴立面 1:100
South elevation



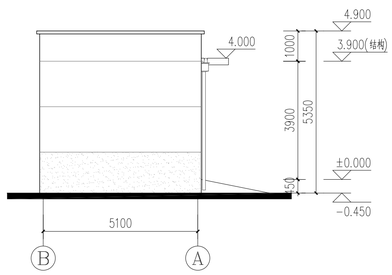

Ⓐ-Ⓑ轴立面 1:100
West elevation




1-1轴立面 1:100
1-1 section




②-①轴立面 1:100
North elevation




Ⓑ-Ⓐ轴立面 1:100
East elevation

浅灰色涂料

深灰色面砖勒脚

Reference Drawings

| Drwg. No. | Title |
|---|---|
| 30-B003701S-T0248 | 建筑通用详图 General details of the building |

Revisions

| | Name | Date | Notes |
|---|---|---|---|
| A | | 2017.11.30 | 第一版 First Edition |

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拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Construction Drawing

M L H V D C - L C S - E C I - D D - C A - 0 2 1 8 - 0 0 3 B

| | 姓名 Name | 日期 Date |
|---|---|---|
| 设计 Designed | | 2019.09.06 |
| 校核 Checked | | 2019.09.06 |
| 审核 Reviewed | | 2019.09.06 |
| 批准 Approved | | 2019.09.06 |
| 比例 Scale 1:100 | 图号 DWG NO. 30-B003701S-T0218-003 | 版本 REV. B |

专用品库建筑图
Special materials storage

| 专业 Speciality | 会签 Sign | 日期 Date |
|---|---|---|
| 结构 Structure | | 2019.09.06 |
| 电气 Electrical | | 2019.09.06 |
| 暖通 HVAC | | 2019.09.06 |
| 给排水 Water supply | | 2019.09.06 |

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